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Derivations of RC LPF and HPF Transfer Functions

The following section will be used as an exercise deriving the transfer functions of second order RC Low Pass and High Pass filters. The filters that make up the graphical EQ will be a combination of cascaded second order high pass and low pass filters coupled using operational amplifiers as buffers to prevent the first filter from loading the second filter. A filter is a circuit that allows spectral components of a signal within a certain bandwidth to pass to its output while attenuating other components of the signal outside of the bandwidth. If a signal is thought of as a mixture, a filter can be thought of as a means by which to separate desirable components and discard undesirable portions of a signal. An RC filter uses resistors and capacitors to achieve this separation.

The first filter we will look at is the second order low pass filter formed by cascading 2 RC low pass filters:

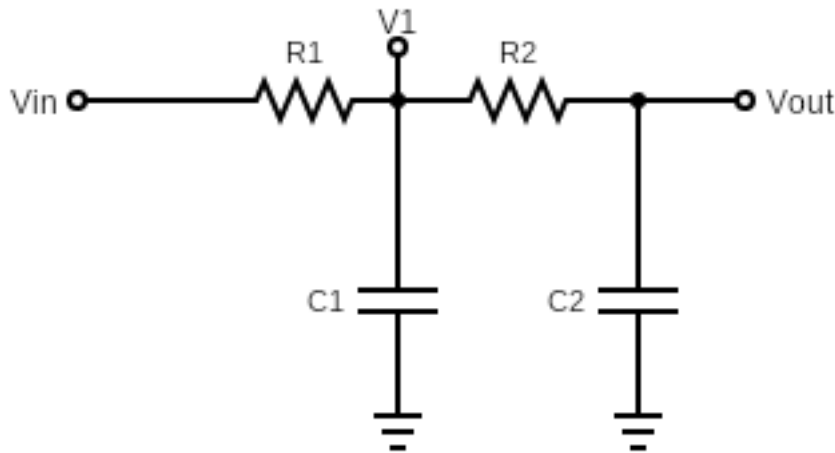


Figure 1. Second Order RC Lowpass Filter

The above circuit, figure 1, is the schematic for the second order low pass filter. The capacitors act as a means to short high frequency signals/signal components to ground while allowing low frequency components to pass to the output. Note that there are 3 nodes of interest in this circuit: The input V_{in} , V_1 , and the output V_{out} . These nodes will be of direct interest for our derivation of the transfer function of this circuit. The resistors R_1 and R_2 , as well as, the capacitors C_1 and C_2 are used to tune the bandwidth and the damping/quality factor of the circuit. Working in the frequency domain will be of interest because it greatly simplifies calculations, turning differential equations and integration into simple algebraic manipulations. In the frequency domain resistors will be unaffected and retain their values (R_1 and R_2 in this case) while capacitors will introduce a pole onto the complex plane with an impedance $\frac{1}{Cs}$, where C is the capacitance of the capacitor and $s = \sigma + j\omega$.

The transfer function of this schematic is of direct interest, not only because it gives us the relationship between the input and output of the circuit, but also because it allows us to estimate the frequency response of the circuit in our design. By setting $s = j\omega$, the frequency response can be estimated for the circuit essentially telling us how the circuit will respond as a function of frequency in radians/second. This is especially important when designing filters because it allows us to estimate the bandwidth of the filter, how steep its roll off is, and whether or not the filter will

sufficiently remove undesired components.

From figure 1, two relationships between voltages in the circuit should be clear when performing KVL and KCL throughout the circuit:

$$V_{out} = V_1 \cdot \frac{\frac{1}{C_2s}}{R_2 + \frac{1}{C_2s}} = V_1 \cdot \frac{1}{R_2C_2s + 1}$$

and,

$$V_1 = V_{in} \cdot \frac{(R_2 + \frac{1}{C_2s}) \parallel \frac{1}{C_1s}}{R_1 + (R_2 + \frac{1}{C_2s}) \parallel \frac{1}{C_1s}}$$

The expression for V_1 can then be further simplified and manipulated,

$$V_1 = V_{in} \cdot \frac{\frac{(R_2 + \frac{1}{C_2s}) \frac{1}{C_1s}}{R_2 + \frac{1}{C_2s} + \frac{1}{C_1s}}}{R_1 + \frac{(R_2 + \frac{1}{C_2s}) \frac{1}{C_1s}}{R_2 + \frac{1}{C_2s} + \frac{1}{C_1s}}}$$

$$V_1 = V_{in} \cdot \frac{(R_2 + \frac{1}{C_2s}) \frac{1}{C_1s}}{R_1(R_2 + \frac{1}{C_2s} + \frac{1}{C_1s}) + R_2 + \frac{1}{C_2s}}$$